

Executive Summary

This audit report was prepared by Quantstamp, the leader in blockchain security.

Type	ERC20 Token	Documentation quality	Medium	
Timeline	2026-02-03 through 2026-02-03	Test quality	Low	
Language	Solidity	Total Findings	1	Fixed: 1
Methods	Architecture Review, Unit Testing, Functional Testing, Computer-Aided Verification, Manual Review	High severity findings ⓘ	0	
Specification	None	Medium severity findings ⓘ	1	Fixed: 1
Source Code	metastreet-labs/usdai-governance-contracts  #2dcc0ac 	Low severity findings ⓘ	0	
Auditors	- Cameron Biniamow Auditing Engineer - Rabib Islam Senior Auditing Engineer	Undetermined severity findings ⓘ	0	
		Informational findings ⓘ	0	

Summary of Findings

The USD.Ai Chip token is an upgradeable ERC-20 token that supports approvals via tokenholder signatures, as well as voting and delegation capabilities. Blacklisted users are restricted from initiating, sending, and receiving token transfers. The blacklist is maintained in an out-of-scope external contract. The bridge admin, a LayerZero OFT adapter contract, executes bridging by burning user tokens when bridged to a remote chain and minting them when bridged to the source chain.

Quantstamp reviewed the `Chip.sol` contract. During the review, the audit team identified one medium-severity issue (**USDAI-1**) in which blacklisted accounts can circumvent the blacklist for voting and delegate their voting power to a non-blacklisted account. Additionally, two auditor suggestions are listed regarding adherence to best practices.

Update: The client fixed **USDAI-1** by restricting blacklisted accounts from delegating their votes and adding a new function `revokeDelegate()`, which is restricted to the `REVOKE_DELEGATE_ADMIN_ROLE`. The auditor suggestions S1 and S2 were acknowledged.

ID	DESCRIPTION	SEVERITY	STATUS
USDAI-1	Blacklisted Accounts Can Delegate Votes	• Medium ⓘ	Fixed

Assessment Breakdown

Quantstamp's objective was to evaluate the repository for security-related issues, code quality, and adherence to specification and best practices.

Disclaimer

Only features that are contained within the repositories at the commit hashes specified on the front page of the report are within the scope of the audit and fix review. All features added in future revisions of the code are excluded from consideration in this report.

Possible issues we looked for included (but are not limited to):

- Transaction-ordering dependence
- Timestamp dependence
- Mishandled exceptions and call stack limits
- Unsafe external calls
- Integer overflow / underflow
- Number rounding errors
- Reentrancy and cross-function vulnerabilities
- Denial of service / logical oversights
- Access control
- Centralization of power
- Business logic contradicting the specification
- Code clones, functionality duplication
- Gas usage
- Arbitrary token minting

Methodology

1. Code review that includes the following
 1. Review of the specifications, sources, and instructions provided to Quantstamp to make sure we understand the size, scope, and functionality of the smart contract.
 2. Manual review of code, which is the process of reading source code line-by-line in an attempt to identify potential vulnerabilities.
 3. Comparison to specification, which is the process of checking whether the code does what the specifications, sources, and instructions provided to Quantstamp describe.
2. Testing and automated analysis that includes the following:
 1. Test coverage analysis, which is the process of determining whether the test cases are actually covering the code and how much code is exercised when we run those test cases.
 2. Symbolic execution, which is analyzing a program to determine what inputs cause each part of a program to execute.
3. Best practices review, which is a review of the smart contracts to improve efficiency, effectiveness, clarity, maintainability, security, and control based on the established industry and academic practices, recommendations, and research.
4. Specific, itemized, and actionable recommendations to help you take steps to secure your smart contracts.

Scope

Files Included

Repo: `https://github.com/metastreet-labs/usdai-governance-contracts`

Included Paths: `src/Chip.sol`

Key Actors And Their Capabilities

- **Default Admin:** The default admin of the `Chip` contract. Assigned during contract initialization. Can grant and revoke the bridge admin role.
- **Bridge Admin:** Granted to a LayerZero OFT adapter contract by the default admin. Controls the bridging of Chip tokens via the `mint()` and `burn()` functions. It is critical that the Bridge Admin is set to the LayerZero OFT adapter contract only. Any non-LayerZero OFT adapter contract bridge admins could burn any user's tokens since `burn()` does not require explicit approval from the token holder.

Findings

USDAI-1 Blacklisted Accounts Can Delegate Votes

• Medium ⓘ Fixed

Update

The client fixed the issue in commit `9052c8e97b13d016dc0e258f3eb6528191ed69f6` by restricting blacklisted accounts from delegating their votes. Additionally, a new function `revokeDelegate()` was added to revoke delegations from accounts before they were blacklisted.

File(s) affected: `src/Chip.sol`

Description: The `Chip` contract restricts blacklisted accounts from initiating, sending, and receiving tokens. However, blacklisted accounts are not restricted from delegating their votes. Depending on how the Governance contract that counts account votes is configured, blacklisted accounts may be prevented from voting directly, but can circumvent this restriction by delegating to a non-blacklisted account.

Recommendation: Since the blacklist state is stored in the external `USDai.sol` contract, the complexity of fixing this issue increases. To isolate the fix solely to the `Chip` contract, consider the following:

1. Override the `VotesUpgradeable._delegate()` function and add a check ensuring the `account` is not blacklisted, preventing the blacklisted account from delegating in the future.
2. Implement a function `revokeDelegate()`, allowing an admin to manually revoke any delegations set by accounts before they were blacklisted.

To restrict blacklisted accounts from delegating their votes atomically, consider:

1. Implement the steps listed in the first fix.
2. Update the `USDai.sol` contract to invoke the newly implemented `Chip.revokeDelegate()` function upon blacklisting an account.

Auditor Suggestions

S1 Improved Error Handling

Acknowledged

Description: In the `mint()` function, when tokens are bridged into the source chain from a remote chain, the bridged tokens are minted, and the `bridged` supply is decreased. However, if the bridged amount exceeds the `bridged` supply, an underflow will occur, and the call will revert.

Recommendation: Consider adding a check to the `mint()` function validating that `amount >= _getSupplyStorage().bridged` and throw an error with a descriptive message if the condition is unmet.

S2 Total Supply Excludes Bridged Tokens

Acknowledged

File(s) affected: `src/Chip.sol`

Description: The total supply of Chip tokens is initially set and minted to the contract admin during initialization. When Chip tokens are bridged to a remote chain, they are burnt, reducing the total supply. This may be misleading to users or other contracts, since the total supply remained the same; only the supply on the source chain decreased.

Recommendation: Consider documenting that the total supply across the source and remote chains is the sum of the total supply and bridged supply.

Definitions

- **High severity** – High-severity issues usually put a large number of users' sensitive information at risk, or are reasonably likely to lead to catastrophic impact for client's reputation or serious financial implications for client and users.
- **Medium severity** – Medium-severity issues tend to put a subset of users' sensitive information at risk, would be detrimental for the client's reputation if exploited, or are reasonably likely to lead to moderate financial impact.
- **Low severity** – The risk is relatively small and could not be exploited on a recurring basis, or is a risk that the client has indicated is low impact in view of the client's business circumstances.
- **Informational** – The issue does not pose an immediate risk, but is relevant to security best practices or Defence in Depth.
- **Undetermined** – The impact of the issue is uncertain.
- **Fixed** – Adjusted program implementation, requirements or constraints to eliminate the risk.
- **Mitigated** – Implemented actions to minimize the impact or likelihood of the risk.
- **Acknowledged** – The issue remains in the code but is a result of an intentional business or design decision. As such, it is supposed to be addressed outside the programmatic means, such as: 1) comments, documentation, README, FAQ; 2) business processes; 3) analyses showing that the issue shall have no negative consequences in practice (e.g., gas analysis, deployment settings).

Appendix

File Signatures

The following are the SHA-256 hashes of the reviewed files. A file with a different SHA-256 hash has been modified, intentionally or otherwise, after the security review. You are cautioned that a different SHA-256 hash could be (but is not necessarily) an indication of a changed condition or potential vulnerability that was not within the scope of the review.

Files

Repo: `https://github.com/metastreet-labs/usdai-governance-contracts`

- `45b...3af./src/Chip.sol`

Test Suite Results

The test suite was run using `forge test`.

```
Ran 1 test for test/ChipUpgrade.t.sol:UpgradeTest
[PASS] test__Upgrade() (gas: 4265248)
Suite result: ok. 1 passed; 0 failed; 0 skipped; finished in 226.54ms (31.45ms CPU time)
```

```
Ran 19 tests for test/StakedChipUpgrade.t.sol:StakedChipUpgradeTest
[PASS] test_AdminCanUpgrade() (gas: 4145608)
[PASS] test_AdminFunctionsWorkAfterUpgrade() (gas: 4197794)
[PASS] test_CanDepositAfterUpgrade() (gas: 4362310)
[PASS] test_CanTransferAfterUpgrade() (gas: 4416201)
[PASS] test_CanWithdrawAfterUpgrade() (gas: 4404903)
[PASS] test_ImplementationVersion() (gas: 4154298)
[PASS] test_MultipleUpgrades() (gas: 12598196)
[PASS] test_NonAdminCannotUpgrade() (gas: 4126189)
[PASS] test_UpgradePreservesBalances() (gas: 4368834)
[PASS] test_UpgradePreservesBridgedSupply() (gas: 4431800)
[PASS] test_UpgradePreservesLockedShares() (gas: 4373355)
[PASS] test_UpgradePreservesPauseState() (gas: 4212913)
[PASS] test_UpgradePreservesRoles() (gas: 4189290)
[PASS] test_UpgradePreservesTotalAssets() (gas: 4479429)
[PASS] test_UpgradeToExtendedImplementation() (gas: 4405466)
[PASS] test_UpgradeWhilePaused() (gas: 4418674)
[PASS] test_UpgradeWithData() (gas: 4149166)
[PASS] test_UpgradeWithLargeBalances() (gas: 4403869)
[PASS] test_UpgradeWithNoDeposits() (gas: 4362632)
Suite result: ok. 19 passed; 0 failed; 0 skipped; finished in 363.31ms (93.56ms CPU time)
```

```
Ran 28 tests for test/StakedChipBridge.t.sol:StakedChipBridgeTest
[PASS] test_AdminCanGrantBridgeRole() (gas: 59138)
[PASS] test_BridgeAdminCanBurn() (gas: 270125)
[PASS] test_BridgeAdminCanMint() (gas: 297299)
[PASS] test_BridgeWithActiveDeposits() (gas: 401993)
[SKIP] test_BridgedSupplyDoesNotAffectSharePrice() (gas: 0)
[PASS] test_BurnDecreasesUserBalance() (gas: 289681)
[PASS] test_BurnEntireBalance() (gas: 273248)
[PASS] test_BurnIncreaseBridgedSupply() (gas: 291541)
[PASS] test_CannotBurnWhenPaused() (gas: 310154)
[PASS] test_CannotMintWhenPaused() (gas: 337733)
[PASS] test_DepositAfterBridgeOperations() (gas: 367037)
[PASS] test_InitialBridgedSupplyIsZero() (gas: 14693)
[PASS] test_InitialTotalShares() (gas: 17488)
[PASS] test_MintDecreasesBridgedSupply() (gas: 319012)
[PASS] test_MintIncreasesRecipientBalance() (gas: 301291)
[PASS] test_MintWithoutPriorBurn() (gas: 97087)
[PASS] test_MultipleBurnsIncrementBridgedSupply() (gas: 314497)
[PASS] test_MultipleUsersBridging() (gas: 440617)
[PASS] test_NonBridgeAdminCannotBurn() (gas: 248847)
[PASS] test_NonBridgeAdminCannotMint() (gas: 27296)
[PASS] test_OnlyBridgeAdminCanBurn() (gas: 257223)
[PASS] test_OnlyBridgeAdminCanMint() (gas: 44782)
[PASS] test_PartialRoundTrip() (gas: 323874)
[PASS] test_RoundTripBurnAndMint() (gas: 304425)
[PASS] test_TotalSharesAfterBurn() (gas: 292495)
[PASS] test_TotalSharesAfterMint() (gas: 320006)
[PASS] test_TotalSharesConsistentAcrossBridgeOperations() (gas: 376583)
[PASS] test_TotalSharesEqualsSupplyPlusBridged() (gas: 246774)
Suite result: ok. 27 passed; 0 failed; 1 skipped; finished in 364.84ms (200.79ms CPU time)
```

```
Ran 49 tests for test/StakedChip.t.sol:StakedChipTest
[PASS] test_AdminCanGrantRoles() (gas: 57802)
[PASS] test_AdminRoleGranted() (gas: 19571)
[PASS] test_CannotDepositToInvalidAddress() (gas: 64482)
[PASS] test_CannotDepositWhenPaused() (gas: 127753)
[PASS] test_CannotDepositZero() (gas: 64283)
[PASS] test_CannotInitializeTwice() (gas: 23861)
[PASS] test_CannotMintToInvalidAddress() (gas: 64805)
```



```
[PASS] test_CannotMintZero() (gas: 65385)
[PASS] test_CannotPauseWithoutRole() (gas: 22696)
[PASS] test_CannotRedeemIfOwnerBlacklisted() (gas: 313381)
[PASS] test_CannotRedeemOnBehalfOfBlacklistedOwner() (gas: 344889)
[PASS] test_CannotRedeemToInvalidAddress() (gas: 258263)
[PASS] test_CannotRedeemZero() (gas: 257991)
[PASS] test_CannotWithdrawIfOwnerBlacklisted() (gas: 316983)
[PASS] test_CannotWithdrawOnBehalfOfBlacklistedOwner() (gas: 354758)
[PASS] test_CannotWithdrawToInvalidAddress() (gas: 257525)
[PASS] test_CannotWithdrawWhenPaused() (gas: 322414)
[PASS] test_CannotWithdrawZero() (gas: 257324)
[PASS] test_ConvertToAssets() (gas: 20093)
[PASS] test_ConvertToShares() (gas: 20655)
[PASS] test_Deposit() (gas: 263057)
[PASS] test_DepositMintsLockedShares() (gas: 254796)
[PASS] test_DepositOnlyMintsLockedSharesOnce() (gas: 318457)
[PASS] test_DepositToReceiver() (gas: 264210)
[PASS] test_FirstDepositorEdgeCase() (gas: 270431)
[PASS] test_ImplementationVersion() (gas: 18961)
[PASS] test_Initialization() (gas: 50486)
[PASS] test_LargeDepositAmount() (gas: 289276)
[PASS] test_MaxDeposit() (gas: 14958)
[PASS] test_MaxMint() (gas: 16154)
[PASS] test_MaxRedeem() (gas: 255590)
[PASS] test_MaxWithdraw() (gas: 259476)
[PASS] test_Mint() (gas: 281465)
[PASS] test_MintMintsLockedShares() (gas: 272460)
[PASS] test_MultipleUsersDepositWithdraw() (gas: 407048)
[PASS] test_NonAdminCannotGrantRoles() (gas: 27487)
[PASS] test_OnlyAdminCanAuthorizeUpgrade() (gas: 4149757)
[PASS] test_PauseUnpause() (gas: 72189)
[PASS] test_PreviewDeposit() (gas: 363046)
[PASS] test_PreviewMint() (gas: 381450)
[PASS] test_PreviewRedeem() (gas: 317354)
[PASS] test_PreviewWithdraw() (gas: 321306)
[PASS] test_Redeem() (gas: 317245)
[PASS] test_RedeemToReceiver() (gas: 324273)
[PASS] test_TotalAssets() (gas: 264216)
[PASS] test_TotalAssetsMultipleDeposits() (gas: 362259)
[PASS] test-Withdraw() (gas: 321708)
[PASS] test-WithdrawToReceiver() (gas: 327995)
[PASS] test-WithdrawWithApproval() (gas: 349573)
Suite result: ok. 49 passed; 0 failed; 0 skipped; finished in 364.93ms (289.83ms CPU time)
```

Ran 36 tests for test/ChipVotes.t.sol:ChipVotesTest

```
[PASS] testFuzz_DelegateAndTransfer(uint256) (runs: 256, μ: 281461, ~: 280722)
[PASS] testFuzz_MultipleDelegations(uint8) (runs: 256, μ: 473493, ~: 387681)
[PASS] test__BlacklistedCannotDelegate() (gas: 69126)
[PASS] test__BlacklistedCannotTransfer() (gas: 70931)
[PASS] test__CannotDelegateToBlacklisted() (gas: 76014)
[PASS] test__ChangeDelegation() (gas: 216688)
[PASS] test__CheckpointsCreatedOnDelegation() (gas: 122163)
[PASS] test__ClockAdvancesWithTime() (gas: 20806)
[PASS] test__ClockModeReturnsTimestamp() (gas: 19427)
[PASS] test__ClockReturnsTimestamp() (gas: 13393)
[PASS] test__DelegateBySig() (gas: 236194)
[PASS] test__DelegateBySigRevertsExpired() (gas: 109480)
[PASS] test__DelegateEmitsEvent() (gas: 125455)
[PASS] test__DelegateToAnother() (gas: 137646)
[PASS] test__DelegateToSelf() (gas: 126841)
[PASS] test__DelegateToZeroAddress() (gas: 143358)
[PASS] test__DelegateVotesChangedEmitsEvent() (gas: 124617)
[PASS] test__DelegateWithZeroBalance() (gas: 81916)
[PASS] test__GetPastTotalSupply() (gas: 25389)
[PASS] test__GetPastTotalSupplyAfterBurn() (gas: 183980)
[PASS] test__GetPastVotesAfterTransfer() (gas: 206018)
[PASS] test__GetPastVotesBasic() (gas: 126811)
[PASS] test__GetPastVotesRevertsForFuture() (gas: 127390)
[PASS] test__MultipleCheckpointsOnTransfers() (gas: 254364)
[PASS] test__MultipleDelegatorsToSameDelegate() (gas: 178416)
[PASS] test__NoDelegateByDefault() (gas: 18909)
[PASS] test__NoVotesWithoutDelegation() (gas: 25789)
```

```
[PASS] test__RevokeDelegate() (gas: 202068)
[PASS] test__RevokeDelegateRequiresBlacklisted() (gas: 75979)
[PASS] test__RevokeDelegateRequiresRole() (gas: 24915)
[PASS] test__SelfDelegateActivatesVotes() (gas: 126634)
[PASS] test__TransferEntireBalance() (gas: 173376)
[PASS] test__TransferFromDelegatedToDelegated() (gas: 280361)
[PASS] test__TransferFromUndelegatedToDelegated() (gas: 185595)
[PASS] test__TransferUpdatesVotes() (gas: 181252)
[PASS] test__TransferWithThirdPartyDelegate() (gas: 178367)
Suite result: ok. 36 passed; 0 failed; 0 skipped; finished in 1.04s (1.13s CPU time)

Ran 5 test suites in 1.10s (2.36s CPU time): 132 tests passed, 0 failed, 1 skipped (133 total tests)
```

Code Coverage

Code coverage for the `Chip.sol` contract is low since the core functionality is untested. The audit team recommends improving the test suite to achieve test coverage above 90%.

Update: Code coverage was significantly improved for the `Chip.sol` contract.

File	% Lines	% Statements	% Branches	% Funcs
src/Chip.sol	73.91% (34/46)	62.50% (25/40)	100.00% (2/2)	80.00% (12/15)
Total	73.91% (34/46)	62.50% (25/40)	100.00% (2/2)	80.00% (12/15)

Changelog

- 2026-02-03 - Initial report
- 2026-02-05 - Final report

About Quantstamp

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Quantstamp’s team consists of cybersecurity experts hailing from globally recognized organizations including Microsoft, AWS, BMW, Meta, and the Ethereum Foundation. Quantstamp engineers hold PhDs or advanced computer science degrees, with decades of combined experience in formal verification, static analysis, blockchain audits, penetration testing, and original leading-edge research.

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Quantstamp’s collaborations and partnerships showcase our commitment to world-class research, development and security. We're honored to work with some of the top names in the industry and proud to secure the future of web3.

Notable Collaborations & Customers:

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- Infra/Staking: TrustWallet, Alchemy, Liquid Collective, Kiln, Galxe, API3, ssv.network, Luganodes
- Immersive: Virtuals Protocol, Wayfinder, OpenSea, Square Enix, Parallel, Camp, Decentraland
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